

usb_joystick_routed

Design Report

Rev 1 | 2026-09-15 | jlcpb-tier1

kicad-tools 0.20.0

Candidate Qualification

The native-consumed clearance repair removes the prior saved/refilled copper divergence and 12 native sliver warnings. Exact recipe source: `bd992dc7572cf7af93be092ded9e259c721ab3e9` , based on factory-rule parent `581087a5` . Saved PCB SHA256: `17d3fc83cb535fcd0ccac2e7b1a6036e272ec5d15a7936400217900b84e20a53` .

This is digital candidate evidence, not supplier approval or physical bring-up. Independent combined-head review and main-target CI remain required before parent integration. Bundled `native-clearance-qualification/provenance.json` identifies the separate recipe, factory checker and corrected readiness verifier.

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	38 (37 SMD, 1 THT, 0 other)
Nets	27
Traces	2050 segments
Vias	153
Board Size	80.0 x 60.0 mm

Stackup

Layer	Type	Thickness (mm)	Material
F.Cu	copper	0.035	--
dielectric 1	prepreg	0.2104	7628
In1.Cu	copper	0.0152	--
dielectric 2	core	1.065	FR4
In2.Cu	copper	0.0152	--
dielectric 3	prepreg	0.2104	7628
B.Cu	copper	0.035	--

Design Overview

Theory of Operation

USB joystick controller — ATmega32U4

USB bus powered, 8MHz crystal, AVR ISP programming

GPIO and ADC map is fixed to Microchip TQFP-44 pinout

Communication Interfaces

Protocol	Signals
SPI	ISP_MISO, ISP_MOSI, ISP_SCK
USB	USB_D+, USB_D-, VBUS

Assembly Notes

1 fine-pitch component

- **Fine-pitch components:** 1 (U1)

Hand-Solder (THT) Components

The following 1 through-hole component is **excluded from the SMT pick-and-place file** and must be hand-soldered (or wave/selective-soldered) after SMT assembly. It appears in the BOM for sourcing.

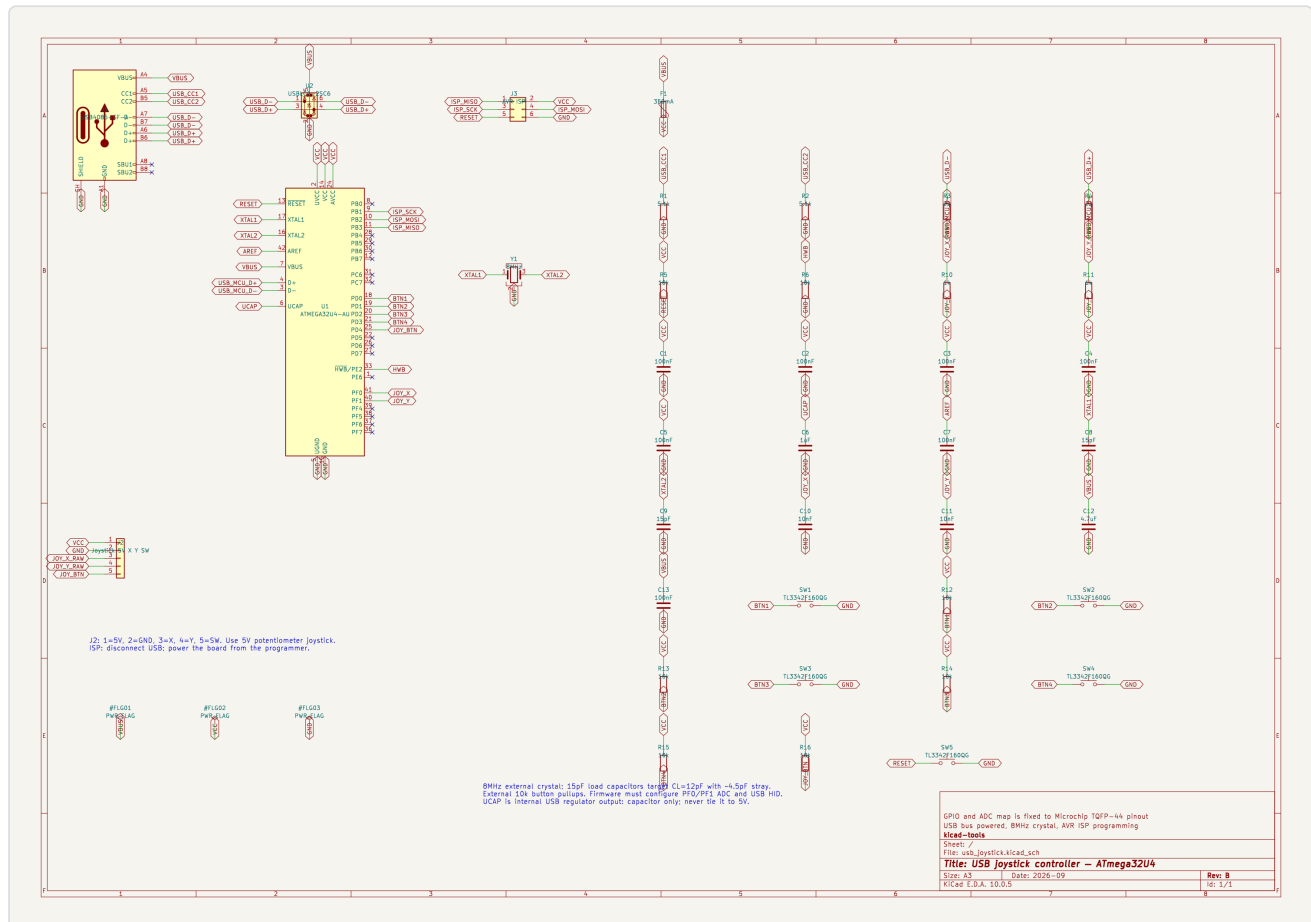
Value	Package	Qty	References
USB4085-GF-A	USB_C_Receptacle_GC T_USB4085	1	J1

ERC Status

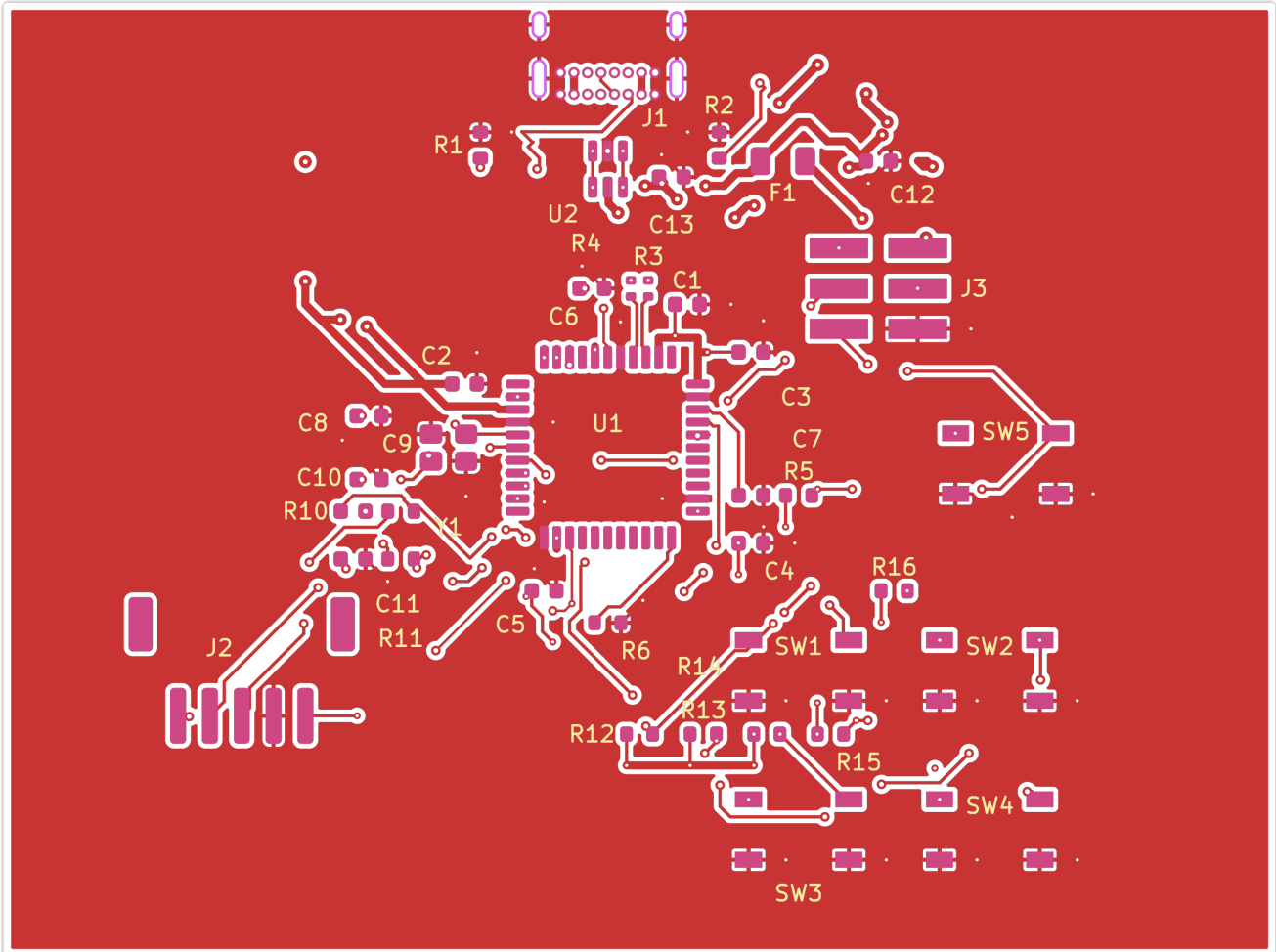
Fresh exact-source recipe native ERC: **0 errors, 0 warnings**. Retained evidence: `native-erc.json`. Exact-source board manufacturing checker independently reports ERC PASSED.

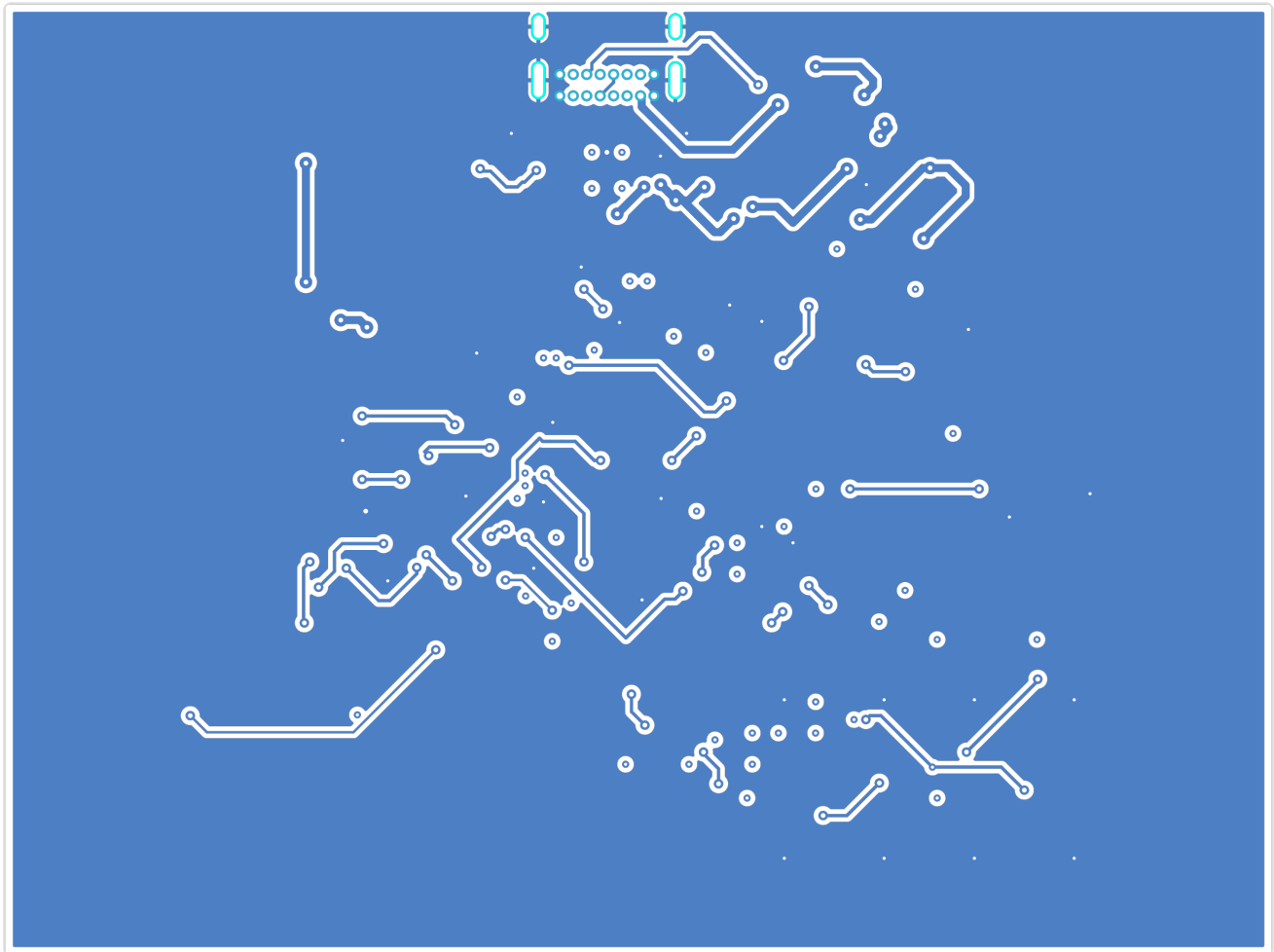
Schematic Overview

Schematic: usb_joystick

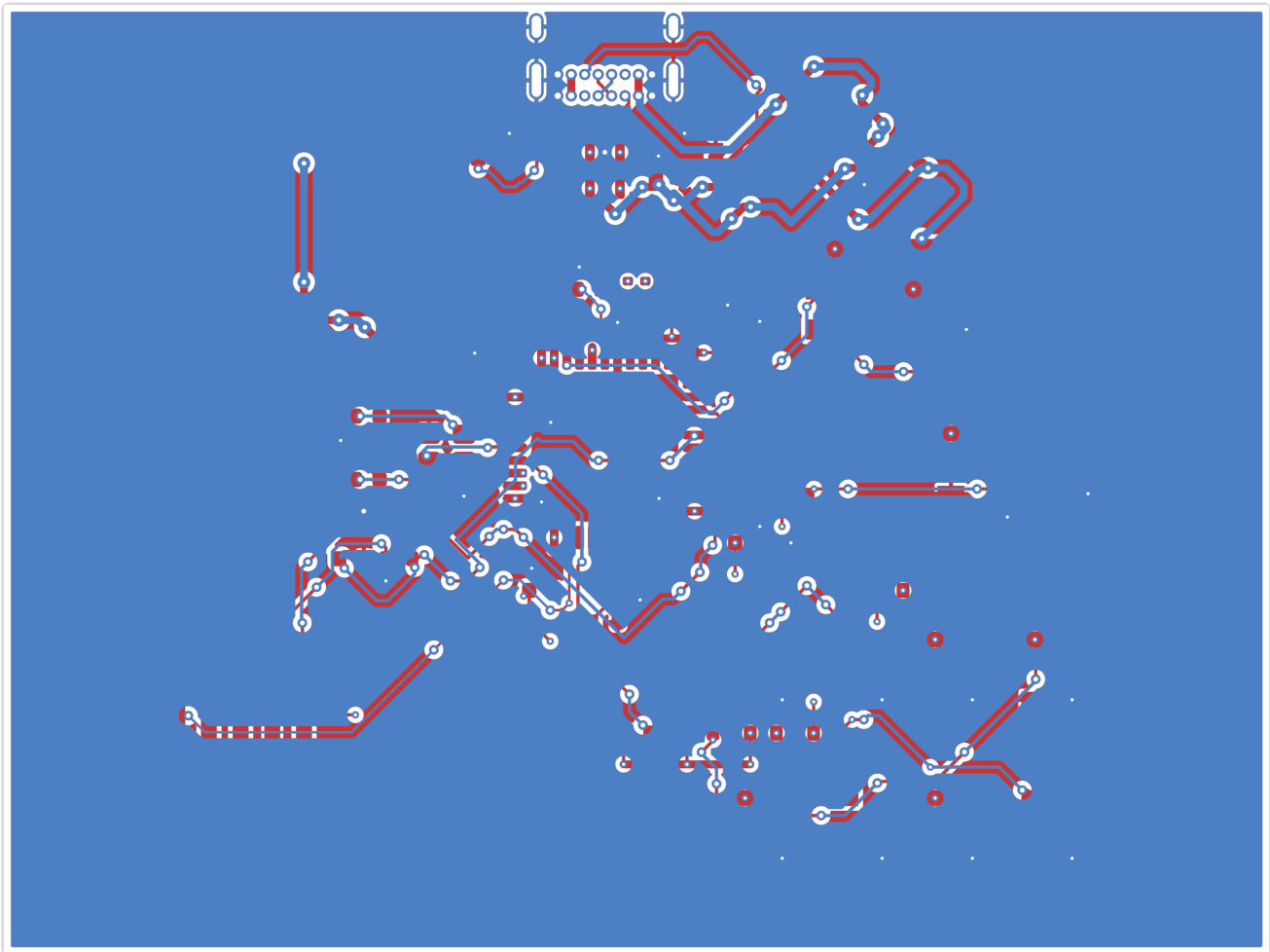


PCB Layout

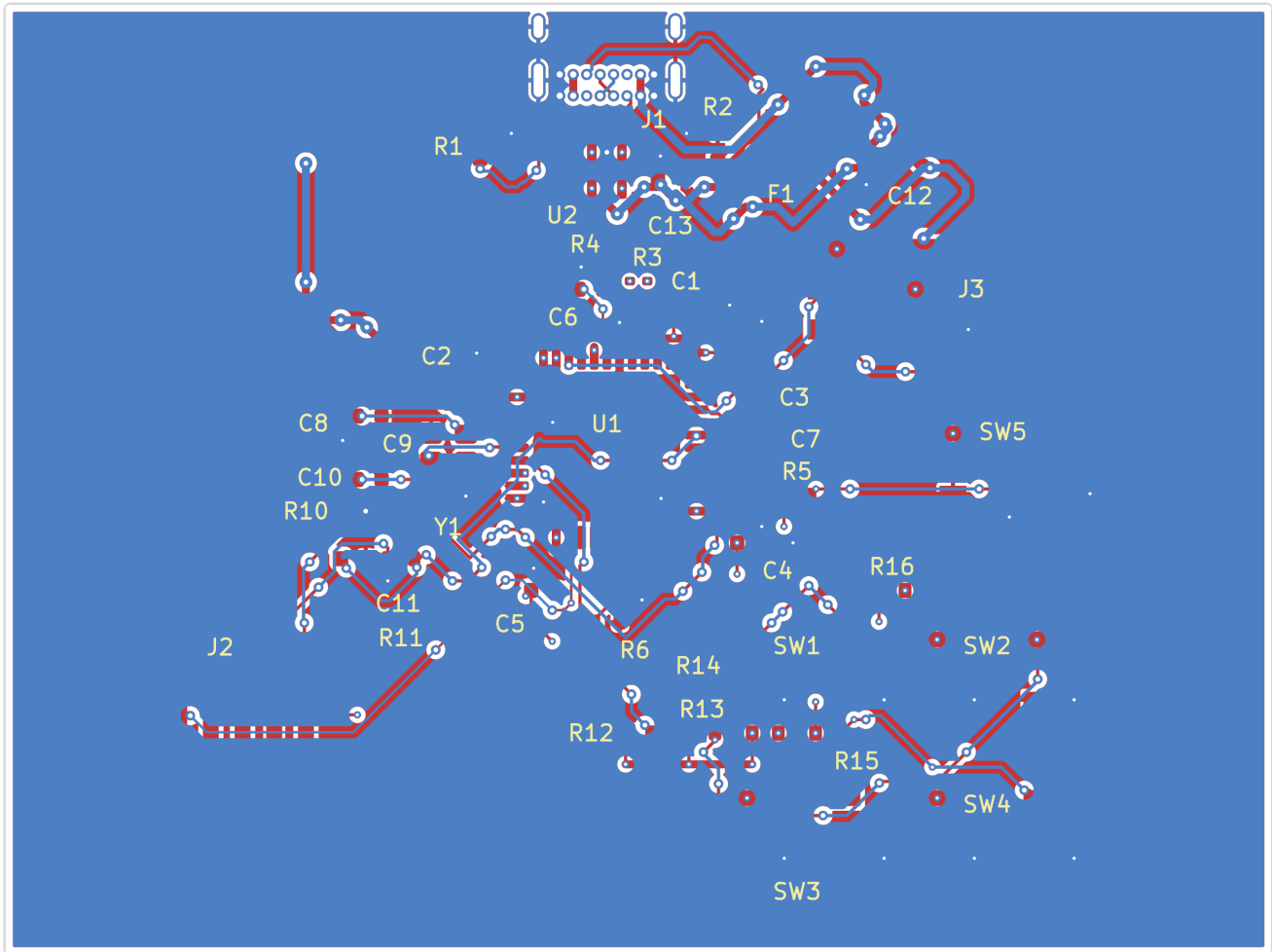




Copper

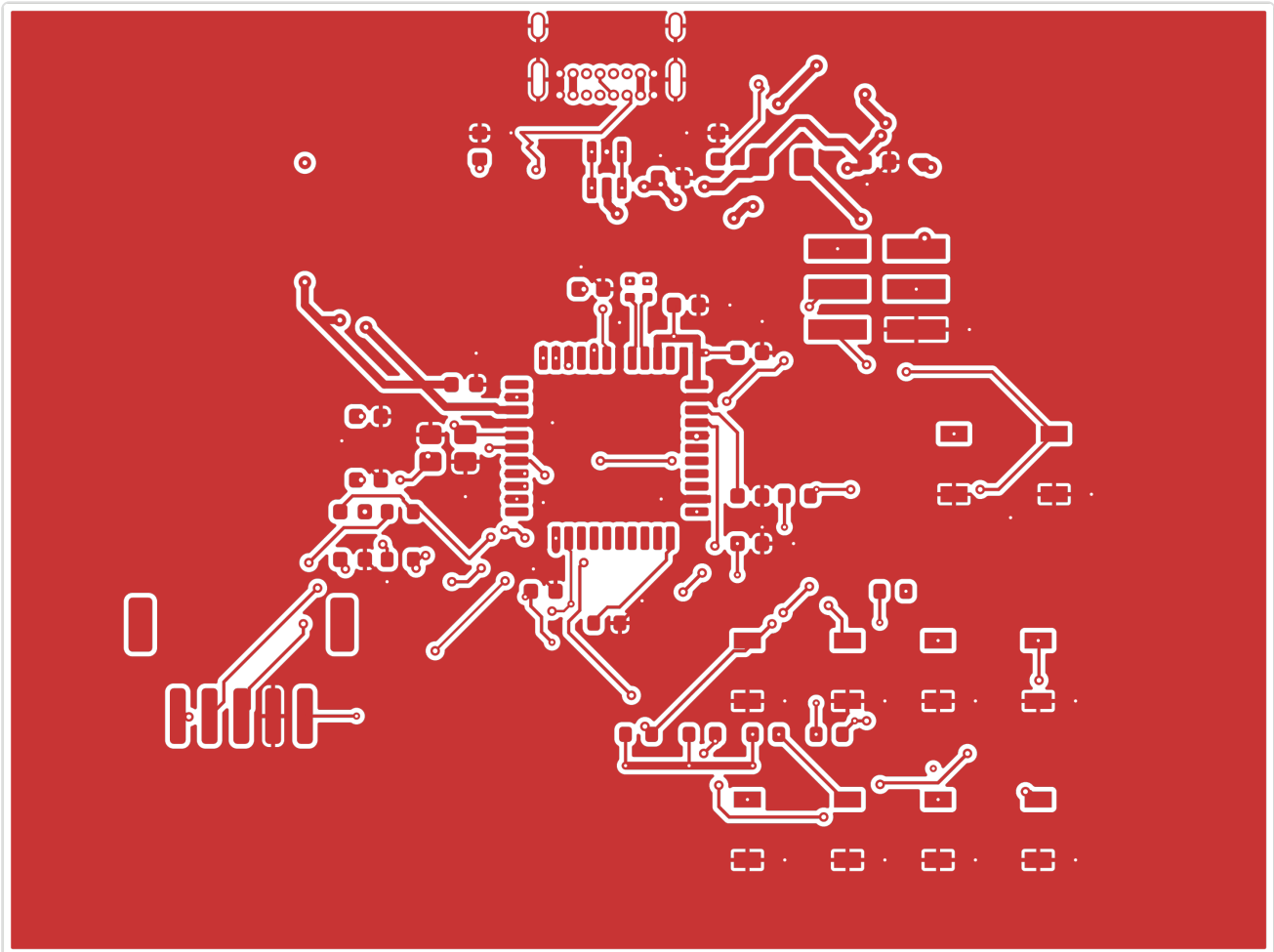


Assembly

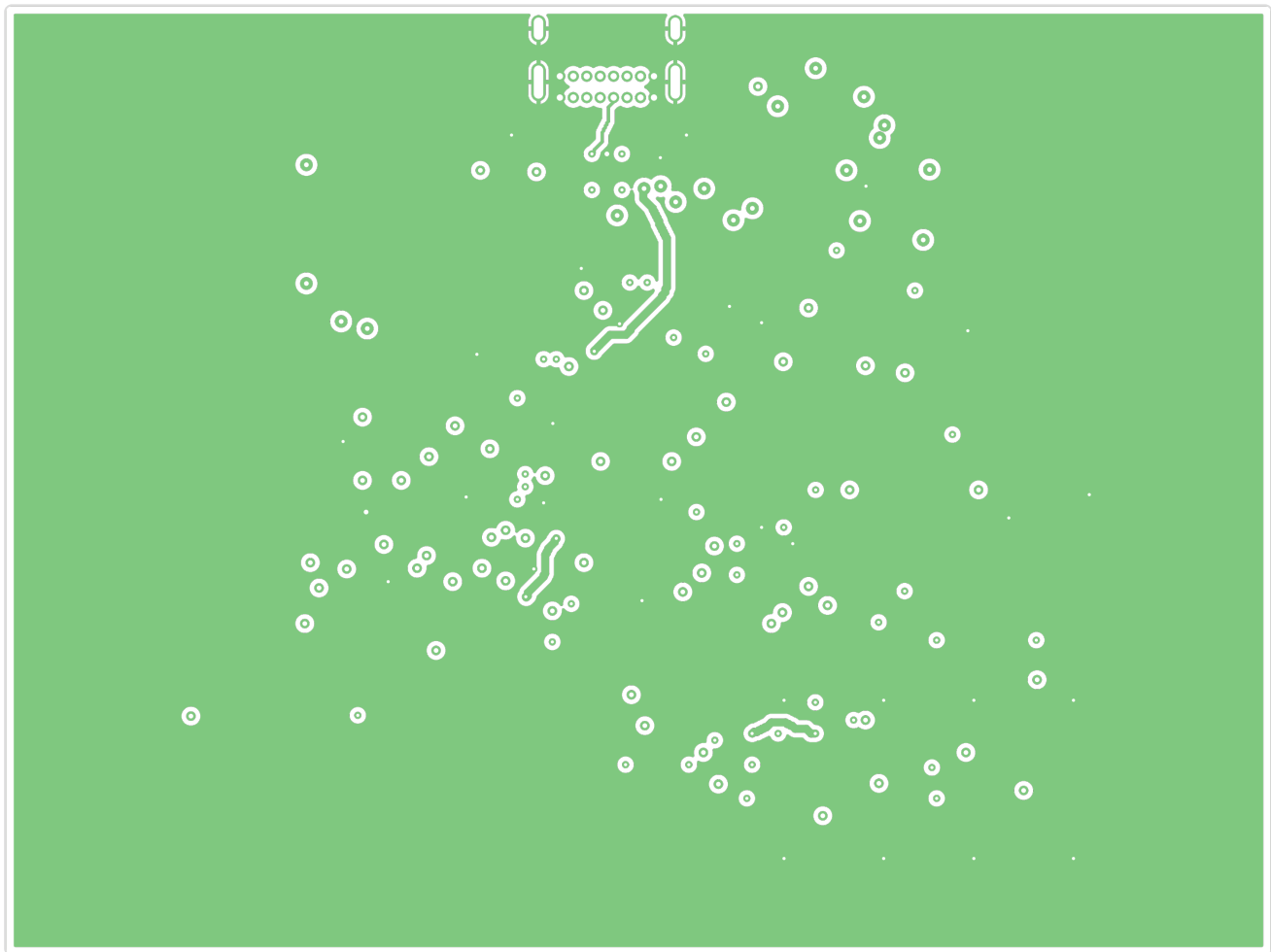


Copper Layers

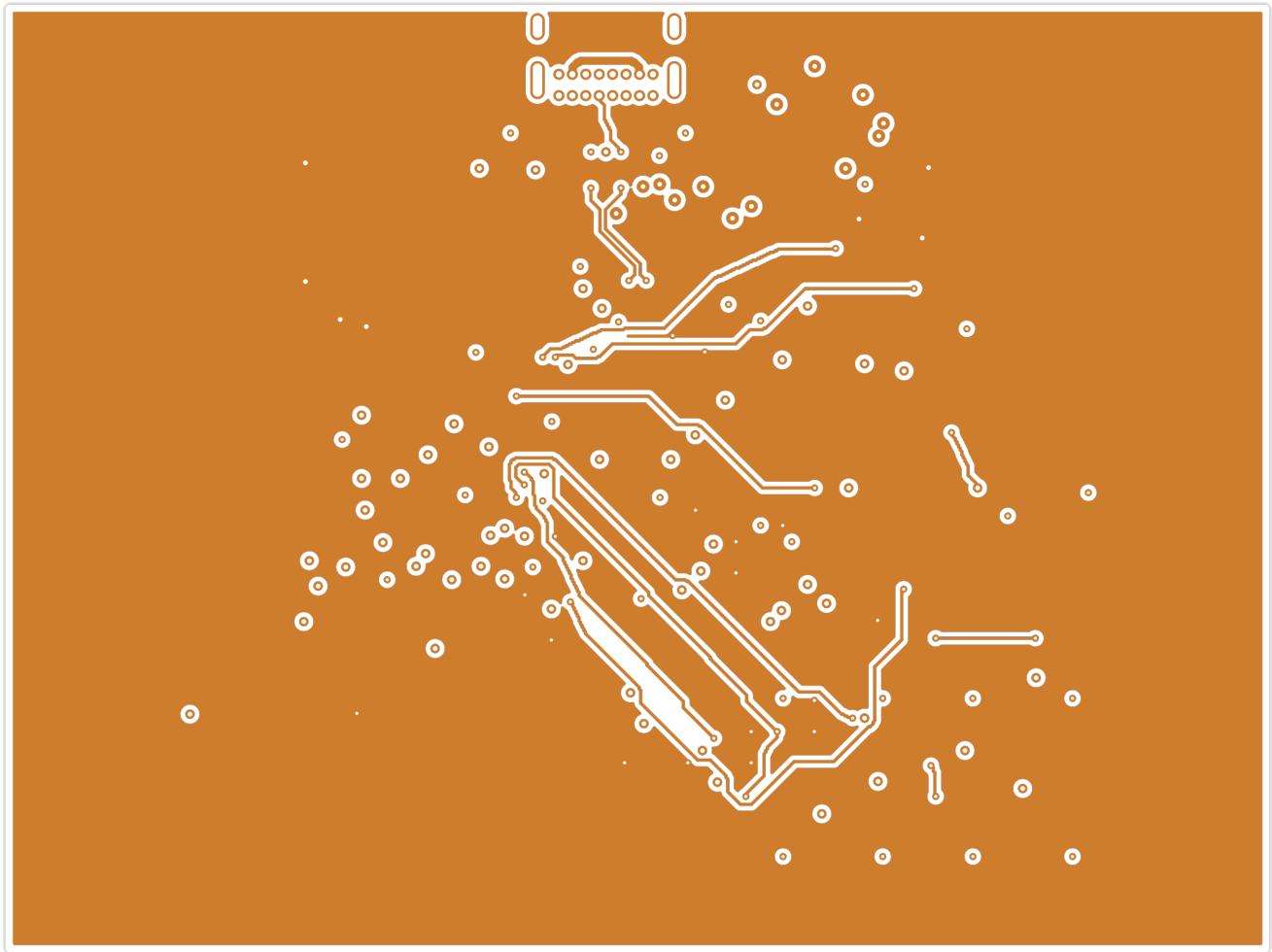
F.Cu



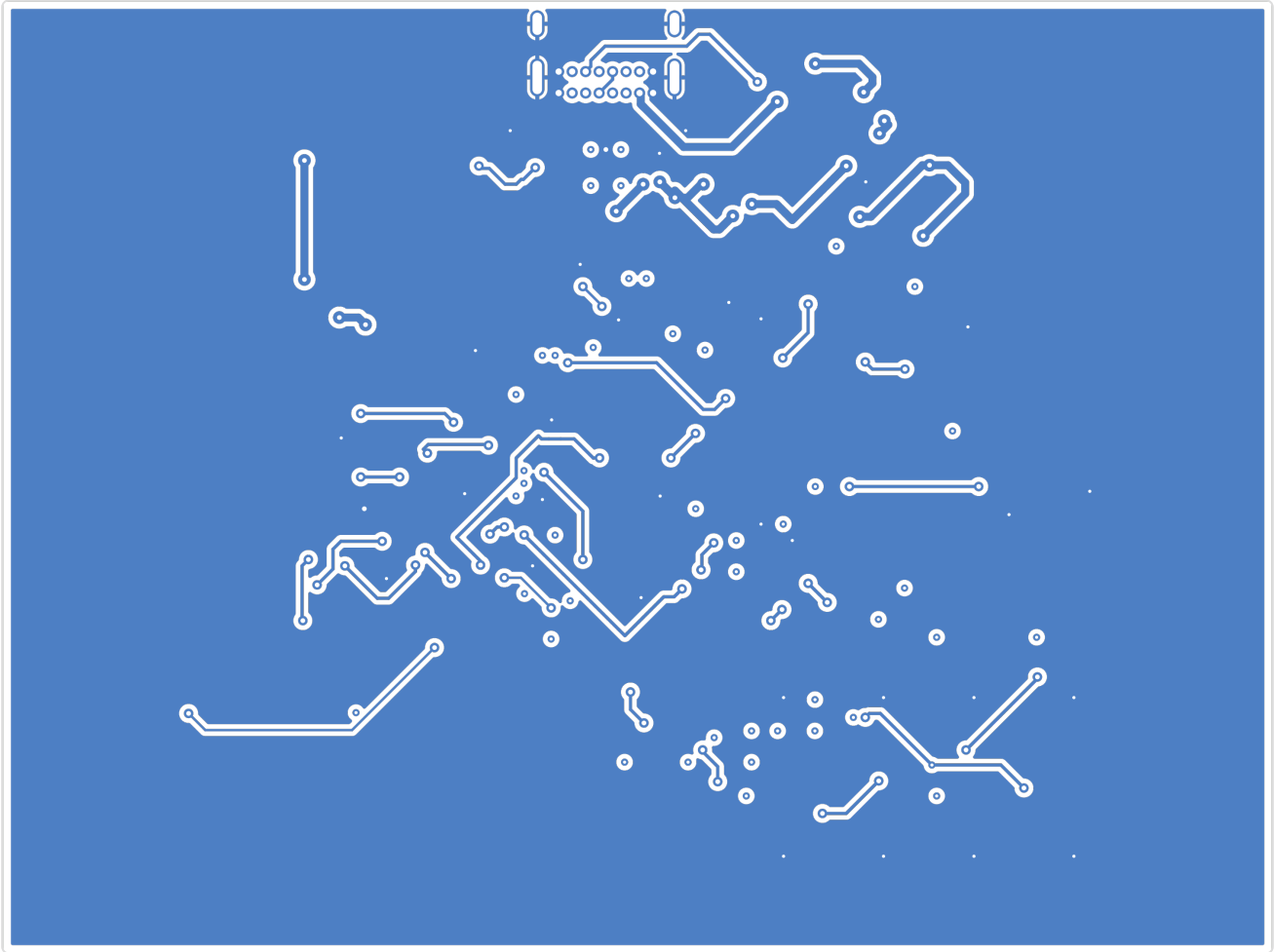
In1.Cu



In2.Cu



B.Cu



Bill of Materials

Value	Package	Qty	References	MPN	LCSC
100nF	C_0603_1608 Metric	7	C1, C2, C3, C4, C5, C7, C13	GRM188R71C 104KA01D	C45000
10nF	C_0603_1608 Metric	2	C10, C11	GRM188R71H 103KA01D	C77053
15pF	C_0603_1608 Metric	2	C8, C9	GRM1885C1H 150JA01D	C71651
1uF	C_0603_1608 Metric	1	C6	GRM188R71A 105KA61D	C97888
4.7uF	C_0603_1608 Metric	1	C12	GRM188R61A 475KE15D	C71633
350mA	Fuse_1206_32 16Metric	1	F1	1206L035/16Y R	C126817
AVR ISP	Samtec_TSM- 103-01-T-DV	1	J3	TSM-103-01-T- DV-P-TR	C21285200
Joystick 5V X Y SW	JST_PH_S5B- PH-SM4- TB_1x05-1MP _P2.00mm_Ho rizontal	1	J2	S5B-PH-SM4- TB(LF)(SN)	C265104
USB4085-GF- A	USB_C_Recep tacle_GCT_US B4085	1	J1	USB4085-GF- A	C7095263
10k	R_0603_1608 Metric	7	R5, R6, R12, R13, R14, R15, R16	RC0603FR-07 10KL	C98220
1k	R_0603_1608 Metric	2	R10, R11	RC0603FR-07 1KL	C22548
22	R_0402_1005 Metric	2	R3, R4	RC0402FR-07 22RL	C114765
5.1k	R_0603_1608 Metric	2	R1, R2	RC0603FR-07 5K1L	C105580
TL3342F160Q G	SW_SPST_TL 3342	5	SW1, SW2, SW3, SW4, SW5	TL3342F160Q G	C2886898

Value	Package	Qty	References	MPN	LCSC
ATMEGA32U4-AU	TQFP-44_10x10mm_P0.8mm	1	U1	ATMEGA32U4-AU	C44854
USBLC6-2SC6	SOT-23-6	1	U2	USBLC6-2SC6	C7519
8MHz	Crystal_SMD_3225-4Pin_3.2x2.5mm	1	Y1	ECS-80-12-33-JGN-TR	C2442665

DRC Status

Pinned native KiCad 10.0.5: **0 errors, 0 warnings, 0 unconnected items**, both on saved copper without refill and on a separate independent native-refilled copy. Exact recipe-source Python factory check: **54 rules, 0 errors, 0 warnings, 0 waived findings**. Strict connectivity covers 27 complete nets; copper LVS is clean over 129 bound pads. Evidence is in `native-clearance-qualification/`.

The earlier export collector reported 14 hole-spacing warnings and optional fill suggestions. Those diagnostics omitted the final scoped factory context; they are superseded by the retained exact-source factory receipt and independent native receipts. No severity, tolerance or allowance was changed. The existing fabrication sidecar and explicit tier remain binding. Optional path-ampacity analysis has no declared current-path sidecar and is not a claimed qualification.

Manufacturing Readiness

Physical candidate gates: **PASSED**. Release review: **PENDING**.

- Saved versus independently refilled copper has **0 mm² symmetric difference on all four layers** (full polygon comparison, not area equality alone).
- Native saved and refilled DRC: **0 findings and 0 opens**. The previous 12 copper slivers are absent.
- The conservative zone-to-foreign-copper target is an explicit native 0.3mm rule, preserving stronger authored rules by refusal. A valid 2.0mm negative rule on unchanged 0.3mm-filled saved copper produces 506 clearance errors; false-condition controls demonstrate rule reachability.
- Factory/process constraints, circuit connectivity, and the immutable historical witness remain preserved. Use the exact four-layer Epoxy-filled & Capped POFV/VIPPO process in the supplied manufacturing requirements.
- Independent exact combined-head review and main-target CI remain required. Supplier approval, assembly inspection, and physical USB qualification remain outstanding.

The readiness verifier's final machine-readable receipt is authoritative for its package checks. It is identified separately from the exact recipe-source factory checker; passing generic main checks cannot substitute for the stronger factory-rule parent.

Routing Status

Metric	Value
Signal Net Completion	100.0% (25/25)
Overall Completion	100.0%
Complete Nets	27 / 27
Zone-Connected Nets	2
Incomplete Nets	0
Unconnected Pads	0

Zone-Connected Nets

- GND
- VCC

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~2.96 USD
Components (estimated)	~2.11 USD
Assembly (estimated)	~2.16 USD
Total (estimated)	~7.23 USD
Batch Quantity	5
Batch Total (estimated)	~36.14 USD